Evaluation

Natural language processing

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Types

- Extrinsic:
  - Embedding LM in an application and measure how much the application improves. Such end-to-end evaluation is called extrinsic evaluation.
- Intrinsic:
  - Train and test
  - Test on unseen samples (test set)

Intrinsic evaluation

- “fit the test set”: Given two probabilistic models, the better model is the one that has a tighter fit to the test data or that better predicts the details of the test data, and hence will assign a higher probability to the test data.
- Training on test set introduces a bias.

Perplexity

- The perplexity (sometimes called PP for short) of a language model on a test set is the inverse probability of the test set, normalized by the number of words. For a test set $W = w_1w_2 \ldots w_N$,:

$$\begin{aligned} \mathrm{PP}(W) &= P(w_1w_2\ldots w_N)^{-\frac{1}{N}} \\ &= \sqrt[N]{\frac{1}{P(w_1w_2\ldots w_N)}} \end{aligned} \tag{3.14}$$

We can use the chain rule to expand the probability of W:

$$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i|w_1\ldots w_{i-1})}} \tag{3.15}$$

Perplexity (Bigram)

$$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i|w_{i-1})}} \tag{3.16}$$

- The higher the conditional probability of the word sequence, the lower the perplexity. In simpler terms:
  - **Low perplexity** = The model is good at guessing the next word (it’s less confused).
  - **High perplexity** = The model struggles to guess the next word (it’s more confused).
- Thus, minimizing perplexity is equivalent to maximizing the test set probability according to the language model.

- Why normalization?

- The intuition behind **normalization in perplexity** is to make the perplexity score independent of the length of the text and provide a **fair comparison** across different models and datasets. Without normalization, longer texts would naturally have higher perplexity scores, even if the model's performance is the same as on shorter texts.
- By normalizing, we focus on the model’s **average performance** over all the words.

Imagine a guessing game:

- Suppose you're playing a word-guessing game with two different sentences. One sentence is 5 words long, and the other is 100 words long.
- If you don't normalize, the score for the longer sentence would be much higher (because there are more chances to make mistakes), even if your "guessing skills" are just as good for both.
- Normalization ensures that we are not just counting how many guesses were made, but rather **how well the model performs on average**, no matter how long the sentence is.

Normalization (Perplexity)

- On average, how confused is the model for each word?" rather than letting longer sentences skew the results. This makes perplexity a **better indicator of overall model performance**.

Branching factor

- **Weighted average branching factor (perplexity)** of a language is the number of possible next words that can follow any word.

Exercise

- Consider the task of recognizing the digits in English (zero, one, two,..., nine), given that each of the 10 digits occurs with equal probability P = $\frac{1}{10}$. Imagine a string of digits of length N.
- Calculate the perplexity:

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$$
\begin{aligned}
\mathrm{PP}(W) &= P(w_1 w_2 \ldots w_N)^{-\frac{1}{N}} \\
&= (\frac{1}{10}^{N})^{-\frac{1}{N}} \\
&= \frac{1}{10}^{-1} \\
&= 10
\end{aligned}
$$

- Suppose that the number zero is really frequent and occurs 10 times more often than other numbers. How it will effect the perplexity?
- Increase or decrease in perplexity?

Solution

- Suppose that the number zero is really frequent and occurs 10 times more often than other numbers. How it will effect the perplexity?
- Consider $p(w_i|w_{i-1})=\frac{10}{19}\ for\ Zero$
- Consider $p(w_i|w_{i-1})=\frac{1}{19}\ for\ 1-9$

- Suppose that the number zero is really frequent and occurs 10 times more often than other numbers. How it will effect the perplexity?
- Suppose the sequence is **zero one two:**
- **PP(zero one two)=p(zero one two)$^{-1/3}$**
- **=(10/19*1/19*1/19)$^{-1/3}$**
- **=**$(\frac{10}{19^3})^{-1/3}$
- **=**$\frac{10^{-1/3}}{19^{-1}} = \frac{19}{10^{1/3}} = 19/$**2.154=8.82**

- Suppose the sequence is **zero zero zero:**
- **PP(zero zero zero)=p(zero zero zero)$^{-1/3}$**
- **=(10/19*10/19*10/19)$^{-1/3}$**
- **=**$\left(\frac{\mathbf{10^3}}{\mathbf{19^3}}\right)^{-1/3}$
- **=**$\frac{\mathbf{10^{-1}}}{\mathbf{19^{-1}}} = \frac{\mathbf{19}}{\mathbf{10}} = \mathbf{1.9}$

- Suppose the sequence is **zero zero one:**
- **PP(zero zero one)=p(zero zero one)$^{-1/3}$**
- **=(10/19*10/19*1/19)$^{-1/3}$**
- **=**$\left(\frac{\mathbf{10^2}}{\mathbf{19^3}}\right)^{-1/3}$
- **=**$\frac{\mathbf{10^{-2/3}}}{\mathbf{19^{-1}}} = \frac{\mathbf{19}}{\mathbf{100^{1/3}}} = \mathbf{19}/4.641$**=4.094**

Entropy

- **Entropy** quantifies the average amount of uncertainty or "surprise" in predicting the next event in a probability distribution. In other words, it tells us how unpredictable a system is based on its probability distribution.

Example

Here's an easy way to understand it:

- If you have a message where every word or symbol is predictable (for example, repeating "a" over and over), the entropy is **low** because there's little surprise.
- But if the message is random, where each word or symbol could be any number of things, the entropy is **high** because it's harder to predict what comes next.

Perplexity is a function of entropy

- Perplexity = 2^H where H is entropy

Formula for Entropy

$$H = -\sum_i p_i \log_2(p_i)$$

Derivation (H)

Probability of an Event

- Let’s say you have a set of possible outcomes or symbols, and each outcome x_i has a probability $p(x_i)$of occurring.
- The sum of the probabilities of all outcomes must equal 1:

$$\sum_i p(x_i) = 1$$

- For example, in a language model, the next word could be one of many, and each word has some probability.

Uncertainty for a Single Event

- The uncertainty (or surprise) of seeing a particular outcome x_i is higher if its probability is lower. This is typically measured as $-\log_2(p(x_i))$.

Expected Uncertainty (Entropy)

- The total entropy is the **expected value** of the uncertainty over all possible outcomes. To find the average uncertainty, we multiply each event's uncertainty by its probability:

$$H = -\sum_i p_i \log_2(p_i)$$

- This equation tells us the **average uncertainty** across all outcomes, weighted by how likely each outcome is.

Self Study (important)

Entropy

- Boston University notes: https://www.cs.bu.edu/fac/snyder/cs505/PerplexityPosts.pdf

Relation between entropy and perplexity

- **Entropy is a measure of uncertainty:**
- **Entropy** quantifies the average amount of uncertainty or "surprise" in predicting the next event in a probability distribution. In other words, it tells us how unpredictable a system is based on its probability distribution.
- When we measure perplexity, we are essentially measuring **how uncertain or confused** the model is when trying to predict the next word, number, or symbol in a sequence. If the model is confident in its predictions (i.e., there's less uncertainty), entropy will be lower, and so will perplexity.

$$H = -\sum_{i} p_i \log_2(p_i)$$

- Suppose that the number zero is really frequent and occurs 10 times more often than other numbers. How it will effect the perplexity?
- Solution: use perplexity defined in terms of entropy to calculate.

1. **Set up the problem:**
    - There are 10 possible numbers (0 through 9).
    - The number 0 occurs 10 times more frequently than each of the other numbers.
    - The probability distribution reflects this skewed frequency.

2. **Assign probabilities:**

   - Let the probability of each of the numbers from 1 to 9 be p.
   - Since 0 occurs 10 times more often than the others, its probability will be $10p$.
   - The total probability across all outcomes must sum to 1:

$$10p + 9p = 1$$

$$19p = 1$$

$$p = \frac{1}{19}$$

   Therefore:

   - The probability of 0 is $10p = \frac{10}{19}$.
   - The probability of each of the numbers 1 through 9 is $p = \frac{1}{19}$.

3. **Calculate the perplexity:** Perplexity is the exponential of the **entropy** of the probability distribution. Entropy H is given by:

$$H = -\sum_i p_i \log_2(p_i)$$

Where p_i are the probabilities of each possible outcome. In our case, we have:

$$H = -\left(\frac{10}{19} \log_2 \frac{10}{19} + 9 \times \frac{1}{19} \log_2 \frac{1}{19}\right)$$

5. **Substitute the values and compute:**

   Now, let's calculate the terms:

$$H = -\left(\frac{10}{19}\log_2\frac{10}{19} + 9 \times \frac{1}{19}\log_2\frac{1}{19}\right)$$

   Using logarithms:

$$\log_2\frac{10}{19} \approx -0.926$$

$$\log_2\frac{1}{19} \approx -4.247$$

4. **Break it down:**

- For the number 0:

$$-\frac{10}{19} \log_2 \frac{10}{19}$$

- For each of the numbers 1 through 9:

$$9 \times \left(-\frac{1}{19} \log_2 \frac{1}{19}\right)$$

Now substitute:

$$H \approx -\left(\frac{10}{19} \times (-0.926) + 9 \times \frac{1}{19} \times (-4.247)\right)$$

$$H \approx -(-0.487 + (-2.011))$$

$$H \approx 2.498$$

6. **Calculate perplexity:** Finally, perplexity is:

$$\text{Perplexity} = 2^H = 2^{2.498} \approx 5.66$$

- **Perplexity is a function of entropy:**
- **Perplexity** is defined as the **exponentiation of the entropy**. It’s a way to turn the concept of entropy (measured in bits or nats) into a more intuitive and interpretable quantity.
- In simple terms, **perplexity is the effective number of choices** the model has to make when predicting the next symbol. This idea of choices stems from the amount of uncertainty or entropy in the distribution of possible next symbols.
  - For example, if the entropy is 3, the perplexity will be 23=82^3 = 823=8, meaning that on average, the model has to choose between 8 possibilities (in the sense of how uncertain it is).

- **Relation between entropy and perplexity:**
- Perplexity=2^H
- Where H is the entropy. This formula tells us that perplexity grows exponentially with entropy. The higher the entropy, the more perplexed the model is.

- **Why entropy helps explain the weighted branching factor:**
- In the specific problem you asked about, where the number 0 occurs more frequently, the probability distribution is **skewed** (non-uniform). Entropy helps us capture this skewness. Since 0 appears more often, the uncertainty (entropy) decreases, and consequently, the **perplexity** decreases, even though the branching factor is still technically 10. Using entropy in this case allows us to account for the fact that the model is less "confused" by the frequent occurrence of 0.

Summary:

- **Entropy** measures how unpredictable a system is.
- **Perplexity** translates this unpredictability into a more interpretable measure that reflects how many effective choices the model has.
- When a distribution is skewed (as in the case where 0 occurs more frequently), entropy—and thus perplexity—decreases, indicating that the model is less "perplexed" and can make better predictions.
- Using entropy to calculate perplexity captures the essence of how well the model predicts outcomes based on the underlying probability distribution.